Debdip Choudhuri

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.52
Adamas International School	Class XII (ISC)	2021	89.60%
St. Stephen's School	Class X (ICSE)	2019	90.20%

MTECH CAPSTONE PROJECT

NS3-Teleop Transport Layer Protocol Analysis

- Worked in **NeWS Lab** under **Dr. Bheemarjuna Reddy Tamma**; built a **3GPP-compliant 5G-NR testbed** in **ns-3/5G-LENA** to quantify **TCP vs. UDP** impact on **XR robot teleoperation latency**, running **270+ seeded Monte Carlo trials** across static, mobile, and handover scenarios for publication-grade protocol-selection evidence.
- Developed **synthetic teleop traffic (240 Hz/70 B commands, 60 Hz/300 B state, 20 fps video)** and benchmarked **motion-to-photon latency across numerologies 0-2**; traced a **non-intuitive latency hump at =1** to **RLC segmentation and buffer dynamics**.
- Extended the framework with a **30 km/h circular mobility model (RLC AM/UM)** and diagnosed a **counter-intuitive UDP underperformance at higher numerologies**, producing a **loss/latency trade-off table**.
- Designed a **4-gNB, 140 m handover topology**; identified and **fixed four latent bugs** (**bearing angle, boresight alignment, beamforming, frozen channel-condition state**) that were silently corrupting handover fidelity, delivering **corrected TCP-vs-UDP benchmarks**.

SELF PROJECTS

Shopping Assistant: Chrome Extension + Local RAG Backend

- **Built** a Chrome extension with a local RAG backend that answers questions about the product on screen and **compares every viewed product** across Amazon and Flipkart, turning scattered listings into **cited, verifiable answers**.
- **Engineered** section-aware retrieval that chunks each product into specs, offers and reviews with metadata-filtered search, plus **balanced per-product retrieval** so no single product crowds out a comparison, and **clickable citations** that highlight the source passage on the page.
- **Benchmarked** answer quality on a 14-question evaluation set: switching from Llama 3.1 8B to Qwen3-235B **raised accuracy from 4/9 to 9/9** on the same questions, reaching **12/14 overall with zero invented facts**, while cheapest-provider routing **cut token prices up to 9x**.

5G Handover Simulation in NS3

- **Engineered a 5G NR handover framework in ns-3.48/5G-LENA**, benchmarking **3GPP A3 vs. speed-adaptive and ML-driven** algorithms (30-120 km/h, 4-gNB, **Gauss-Markov**) via multi-seed sweeps and paired statistical testing **delivered a 32% failure-rate reduction at vehicular speeds**.
- **Designed a speed-tiered handover algorithm** that scales A3 hysteresis and time-to-trigger with UE velocity. It **cut handover failure rate by 32% (22.8% to 15.4%) at 120 km/h** ($p<0.05$, n=15 seeds) with no loss in throughput or latency.
- **Found and patched 3 crash bugs in the 5G-LENA NR module** (an X2 handover null-pointer bug and 2 stale-state assertions). They were aborting 31% of runs; after the fixes, 45/45 runs completed cleanly.

Image Classification Model (Deep Learning Course Project)

- **Built** a from-scratch CNN classifier (no pretrained weights) for a CLEVR-style binary vision task on 18K images, **reframing** it as object-presence detection and **achieving** ~0.78 leaderboard accuracy under strict competition constraints.
- **Designed** a modified ResNet-18 with a custom Global Sum Pooling layer and two-layer MLP head that **preserves** localized object cues average pooling dilutes, **outperforming** deeper, heavier baseline architectures.
- **Engineered** a reproducible 4-fold stratified cross-validation pipeline with AdamW, cosine LR scheduling, mixed-precision training, and early stopping, then **combined** all folds via soft-voting ensembling to **reduce** prediction variance.

TECHNICAL SKILLS

- **Programming and Scripting Languages** : C/C++, Python, SQL.
- **Software Development Tools / Platforms** : NS3, Docker, WireShark, Visual Studio Code, Jupyter Notebook.
- **Data Science and Machine Learning Tools / Libraries** : RAG, LangChain, Pandas, NumPy, Matplotlib, CNN, RNN, LSTM, Transformers, Deep Learning.

POSITIONS OF RESPONSIBILITY & ACHIEVEMENTS

- **Teaching Assistant Advanced Computer Networks** : Designed and graded two programming assignments: a Mini-FTP implementation over TCP and UDP (with stop-and-wait reliability, retransmission, and MD5 integrity checks) and a Wireshark-based packet capture and analysis lab covering HTTP, TCP/UDP, and traceroute.
- **Graduate Aptitude Test in Engineering(GATE) CSE 2025** qualified.
- **West Bengal Joint Entrance Examination(WBJEE) 2021** qualified.

RELEVANT COURSES

Advanced Computer Networks, Deep Learning, Foundations of Machine Learning, Network and Wireless Systems, Network Security, Data Structures and Algorithms.

HOBBIES

Playing Table Tennis, Football, watching Formula 1.